Medicare Chronic Conditions Cost Analysis and Network Optimization

Teams and Roles

Data Scientist Interview Candidate: Miao Wei

Business Interviewer

Data Scientist Interviewer

Terms

1. Medicare: US federal health insurance for individuals aged 65 and older
2. Provider: Physicians or institution treating patients, identified by NPI (national Provider Identifier)
3. Chronic Conditions: Long-term health conditions like diabetes, CHF, COPD

Project Background

1. Raw Requirement

The UHC Network Affordability team is focused on getting our members the right care for the best possible price in the most convenient manner.

The task at hand is to perform an exploratory **data analysis (EDA)** on publicly available claim data for **Medicare members** in the US. **Medicare is the government healthcare program for US citizens aged 65+**.

The aim is to understand **how the cost of treating certain chronic conditions varies across different providers**.

2. Business Understanding

Medicare members' chronic treatments involve varying costs across different providers. By analyzing these patterns, we can optimize UHC's healthcare network for better affordability and care. Specifically, analyzing chronic condition costs allows identification of high-cost treatment pathways and outlier providers. By benchmarking these costs across the network, we can recommend adjustments such as promoting more efficient providers, negotiating better rates, designing targeted intervention programs for high-risk members, and ultimately improving both cost efficiency and patient outcomes.

User Case Example

Consider a Medicare patient diagnosed with diabetes. This patient might receive treatment from several different health provider, with each provider incurring different levels of cost for similar services.

For example,

Provider A charge an average of $500 per diabetic patient annually for regular monitoring and basic care.

At the same time, Provider B charges $1200 for similar services without significant improvements in outcomes. By analyzing treatment cost data for diabetic patients across various providers, we can identify that provider A offers better cost efficiency. This insights allow pre-pay team to guide diabetic patients toward more cost-effective providers like provider A. So that the patient can get a better patient outcomes and reduced overall healthcare expenditure across the network.

Project Scope

1. Public CMS Medicare data;
2. Focus on the chronic illnesses and treatment costs;
3. Benchmark provider performance based on cost;

High-Level Goals

1. Identify patterns in chronic disease treatment costs;
2. Benchmark provider costs;
3. Recommend optimizations to the healthcare network;

Specific Research Questions (Low-Level Goals)

1. Basic Summaries
    a. What is the distribution of races?
    b. What is the most common chronic illness combination?
    c. Which chronic illness combination has the total hight cost?
    d. Which chronic illness combination has the highest cost per member?
2. Benchmarking

    The aim is to understand the distribution of cost across providers treating members with these chronic illnesses. Benchmarking providers across types of care is often a helpful starting point to begin working with areas of excessive cost.

   a. For each provider (use AT_PHYSN_NPI) & chronic illness, calculate the cost per member.
   b. For each chronic illness combination, represent the distribution of costs per provider.
   c. How does this change if we filter out cases where a given chronic illness & provider NPI combination only has one member?
   d. Which providers are consistently expensive across chronic illnesses they treat?

Data Features

The data dictionary has attached in the appendix.

Note:

1. Illnesses value range: [1,2] which 1 means Yes, 2 means No
2. BENE_RACE_CD: range: [0,1,2,3,4,5,6];

Data Analysis

The beneficiary summary file has several chronic illness columns for each member. These are Boolean fields.

· Convert these columns into a single categorical variable, concatenating multiple true diagnoses.

· If a member has three or more chronic conditions, categorise these as "Multiple".

· Join claims & benefit data.

Two aspects:

1. Basic Summary
   a. Descriptive statistics.
2. Benchmarking
   a. Group-by aggregation on chronic conditions and provider NPIs.
   b. Cost distribution visualization.
   c. Filtering for robust sample sizes.

Proposal Solutions

- Benchmark providers based on average cost per condition;
- Exclude low sample outliers;
- Identify target providers for network adjustments.
- Apply Clustering to segment providers.
  - K-means;
  - Hierarchical Clustering;
  - Others
- Apply AI-based analytics to automate cost anomaly detection:
  - Isolation forest to detecting high-cost providers or patients;
  - Z-score statistical Method to identifying extreme outliers;
  - DBSCAN;
- Apply predictive models:
  - XGboost: to predict future high-cost patients;
  - Linear Regression: to estimating future provider treatment costs;

Expected Outcomes

- A ranked list of chronic condition combinations by cost;
- Distribution profiles of provider's cost;
- Top K of consistently high-cost providers;
- Predictive models to support future network optimization;

Evaluation Methods

- Compare average and median costs per chronic condition;
- Visualize cost distributions;
- SME verify;
- Evaluate ML model performance with confusion metrics;

Experiment Steps

1. Data Collection;
2. Data Preprocessing;
3. Feature Engineering for Chronic conditions;
4. Basic EDA;
5. Benchmarking provider costs;
6. Apply clustering anomaly detection, and regression models;
7. Visualize results;

Tech and Tools

1. Pandas;
2. Matplotlib or other visualization tools, Tableau?;
3. Numpy;
4. Sklearn;

Risk Assessment

1. Provider with insufficient sample size may skew results;
2. Public dataset may not capture recent trends;

Communication and Timeline

Data collection: 27th April

Data analysis: 27th April

Data processing: 27th April

Basic Analysis: 27th April

Model training: 28th April

Model testing: 28th April

Timeline: by 10am Monday 27th April (Confirmed with Business team in email, 28th)

Future Works

- Add quality of care measures, not just costs;
- Extend analysis to inpatient claims;
- Predict future high-cost patients using machine learning and neural network.
- Apply the LLM to verbally describe the output to business users;
- Deploy AI tool for real-time network monitoring;
- Construct the knowledge graph and external knowledge base;

Questions

To confirm with the business team and product team:

1. Should we also consider regional factors?
2. Are we targeting cost reduction, quality improvement?
3. Where should the service be placed? Prepay or Post-pay?

4. How do we validate provider performance beyond costs?

Notes

Chronic condition "None" cases should be excluded in benchmarking.

Appendix

1. Member Benefit Data: https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs/Downloads/DE1_0_2009_Beneficiary_Summary_File_Sample_20.zip
2. Outpatient Claim Data: https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs/Downloads/DE1_0_2008_to_2010_Outpatient_Claims_Sample_20.zip
3. User Documentation: https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs/Downloads/SynPUF_DUG.pdf
4. Additional Info: https://www.cms.gov/Research-Statistics-Data-and-Systems/Downloadable-Public-Use-Files/SynPUFs

Data Dictionary

Beneficiary Table		Patient Table	
DESYNPUF_ID	Member ID	DESYNPUF_ID	Member ID
BENE_BIRTH_DT		CLM_ID	Claim ID
BENE_DEATH_DT		SEGMENT	
BENE_SEX_IDENT_CD		CLM_FROM_DT	
BENE_RACE_CD	Race Code	CLM_THRU_DT	
BENE_ESRD_IND		PRVDR_NUM	
SP_STATE_CODE		CLM_PMT_AMT	Claim Payment Amount
BENE_COUNTY_CD		NCH_PRMRY_PYR_CLM_PD_AMT	
BENE_HI_CVRAGE_TOT_MONS		AT_PHYSN_NPI	Provider ID
BENE_SMI_CVRAGE_TOT_MONS		OP_PHYSN_NPI	

BENE_HMO_CVRAGE_TOT_MONS		OT_PHYSN_NPI	
PLAN_CVRG_MOS_NUM		NCH_BENE_BLOOD_DDCTBL_LBLTY_AM	
SP_ALZHDMTA	Alzheimer or related disorders or senile	ICD9_DGNS_CD_1 --- ICD9_DGNS_CD_10	
SP_CHF	Heart Failure	ICD9_PRCDR_CD_1 --- ICD9_PRCDR_CD_6	
SP_CHRNKIDN	Chronic Kidney Disease	NCH_BENE_PTB_DDCTBL_AMT	
SP_CNCR	Cancer	NCH_BENE_PTB_COINSRNC_AMT	
SP_COPD	Chronic Obstructive Pulmonary Disease	ADMTNG_ICD9_DGNS_CD	
SP_DEPRESSN	Depression	HCPCS_CD_1 --- HCPCS_CD_45	
SP_DIABETES	Diabetes		
SP_ISCHMCHT	Ischemic Heart Disease		
SP_OSTEOPRS	Osteoporosis		
SP_RA_OA	rheumatoid arthritis and osteoarthritis (RA/OA)		
SP_STRKETIA	Stroke/transient Ischemic Attack		
MEDREIMB_IP			
BENRES_IP			
PPPYMT_IP			
MEDREIMB_OP			

BENRES_OP			
PPPYMT_OP			
MEDREIMB_CAR			
BENRES_CAR			
PPPYMT_CAR			